HW7 - Project Charter

Smart Parking Platform | October 29, 2026

1. Project Information

Field	Description
Project Name	Smart Parking Platform
Project Type	Software Development / Academic Project
Project Manager	Alex Carter (fictional)
Methodology	Agile / Scrum
Proposed Sponsor	Parking Facility Management / Investor (fictional)
Status	Academic Planning and Development
Version / Date	1.0 / October 29, 2026

2. Project Purpose

Drivers may lose time searching for parking, creating frustration and congestion. The Smart Parking Platform proposes a digital way to view parking availability and help facility operators manage related information more efficiently.

3. Project Objectives

- Provide a usable interface for locating and viewing available spaces.
- Implement authentication and account-management features.
- Provide backend services and a database for parking information.
- Support basic administrative reporting and analytics.
- Use a scalable architecture and complete academic deliverables on schedule.

4. Project Scope

In scope: registration and login; availability interface; backend and database; administrative tools; reporting; testing and quality assurance; GitHub documentation.

Out of scope: physical construction or sensors; citywide traffic-control integration; payment processing; guaranteed live availability without connected data; multi-site commercial deployment.

5. Major Deliverables

- Project Vision and Scope and Software Requirements Specification
- Agile backlog, project schedule, resource and cost plans
- RACI Matrix, Risk Register, Stakeholder Analysis
- Team Charter and Project Charter
- Functional prototype and weekly stakeholder update videos

6. Project Milestones

Milestone	Phase
Initiation and scope definition	Initiation
Requirements and backlog	Planning
Resource, cost, risk and stakeholder plans	Planning
Interface and backend build	Implementation
Integration and quality testing	Testing
Final demonstration and documentation	Closure

Exact dates are governed by the existing Gantt chart and project schedule.

7. Resources and Budget

The fictional team includes Alex Carter (Project Manager) and Jordan Lee (Software Developer), with other assignments as documented in the HW5 Resource Plan and RACI Matrix. Technical resources include GitHub, development and testing tools, database technologies, and project documentation tools. The budget baseline follows the HW5 Cost Management Plan; significant variances require review.

8. Primary Risks and Responses

Risk	Response
Integration failure	Incremental integration testing
Schedule delay	Review milestones and critical tasks
Inaccurate parking data	Validate inputs and update procedures
Security vulnerability	Secure authentication and security testing
Resource constraints	Prioritize essential functionality
Requirement changes	Review impacts and refine backlog

9. Stakeholders

Stakeholders include the Project Manager, development team, proposed sponsor/investor, parking facility management, drivers/end users, and the University Instructor. Engagement priorities and communication plans are defined in the HW7 Stakeholder Analysis.

10. Success Criteria

- Functional prototype demonstrates planned essential features.
- Users can navigate the interface and access parking availability data.
- Authentication and backend services function as intended.
- Critical defects are resolved before the final demonstration.
- Documentation is complete, accurate, and submitted by academic deadlines.

- Work remains aligned with approved scope and planned resources.

11. Governance and Approval

The Project Manager coordinates execution and assesses proposed changes to scope, cost, schedule, or resources. This charter is an **academic planning document**, not evidence of a real commercial sponsor or external authorization.

Role	Name	Status
Project Manager	Alex Carter (fictional)	Proposed
Project Sponsor	Fictional sponsor	Not formally approved
Academic evaluator	University Instructor	Subject to evaluation

12. Conclusion

This charter defines the proposed Smart Parking Platform's purpose, objectives, boundaries, deliverables, governance, and evaluation criteria. It serves as a high-level reference throughout the academic development lifecycle.